Vidyavardhini's College of Engineering and Technology, Vasai
Department of Computer Science & Engineering (Data Science)

Course: Machine Learning Lab

Course code: CSC604

Year: TE **SEM: VI**

Report on Mini Project
Topic:- Handwritten Digit Recognizer
Group Members Name: Satyam Maurya Shravan Surve Mayank Vartak Tushar Sable
Group Roll Number: 62 67 70 71

Checked by

Name of Faculty : **Nivedha Raut**

Signature :

Date : **10-4-26**

Title of Mini Project: Handwritten Digit Recognizer

Background of technolgy used:

Python — primary language (100%)
PyTorch — deep learning framework for building and training the neural network
Tkinter — GUI framework for the real-time visualizer interface
MNIST dataset — the handwritten digit dataset used for training

Literature Review (Significance of Mini Project):

- MNIST Dataset: 70,000 labeled 28×28 grayscale images; standard benchmark for digit recognition research
- Architectures: Feed-forward neural networks and CNNs are primary approaches; LeNet-5 achieves over 99% accuracy
- Preprocessing: Pixel normalization (0–1 range), image flattening to 784 values, and grayscale conversion
- Training Methods: Cross-entropy loss, ReLU activation functions, softmax output for 10-class classification
- Regularization: Dropout layers (typically 0.5 rate) and weight decay prevent overfitting
- Performance: Ensemble CNNs achieve 0.21% error rate; traditional SVMs achieved 0.8% error
- Real-time Visualization: Network internals can be visualized but requires balancing model complexity with computational efficiency

Comment on Loss Function Minimization and Generated Output:

- Loss Function: Cross-entropy loss measures the difference between predicted and actual output; minimized through backpropagation and gradient descent
- Optimization: Adam and Adadelta optimizers adjust weights iteratively to reduce prediction error across training epochs
- Output Layer: 10 neurons (one per digit 0–9) with softmax activation convert raw outputs into probability distributions
- Prediction: The network selects the digit with the highest probability; confidence levels for each class (0–9) are displayed in real-time
- Convergence: Training monitors validation accuracy to ensure the model converges properly and generalizes to unseen data
- Error Reduction: Effective minimization results in 99%+ accuracy on test sets; poor minimization indicates overfitting or insufficient model capacity

GitHub Link of Code:

https://github.com/satyamm14/digit-recognition-w-realtime-nn-visualizer

Conclusion:

Cross-entropy loss measures the difference between predicted and actual output, minimized through backpropagation and gradient descent. Adam and Adadelta optimizers adjust weights iteratively to reduce prediction error across training epochs. The output layer contains 10 neurons (one per digit 0–9) with softmax activation that converts raw outputs into probability distributions. The network predicts by selecting the digit with the highest probability, displaying confidence levels for each class (0–9) in real-time. Training monitors validation accuracy to ensure the model converges properly and generalizes to unseen data. Effective loss minimization results in 99%+ accuracy on test sets, while poor minimization indicates overfitting or insufficient model capacity.